

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Observations are represented by positive joint kernels and Rokhlin disintegrations; geometric currents are used only to calculate coarea densities. The exact information state is the full unnormalized belief on an oriented blow-up of the positive cone, where distinct zero-evidence directions are retained rather than simultaneously collapsed and used. A regular positive posterior kernel is Feller on this compact state. An inserted exact-number coefficient controls additive path/contact observations. Growing finite nets of bounded-Lipschitz tests, rather than a fixed list of moments, approximate the full belief game with an explicit Bellman error, and a fully specified regular parametric class satisfies a controlled Bernstein–von Mises theorem.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: POSITIVE DISINTEGRATIONS, BLOWN-UP EVIDENCE, AND GROWING BELIEF COORDINATES

Posterior probabilities and geometric currents are different objects. Observations are represented by a positive joint kernel and Rokhlin disintegration; currents are used only to calculate regular coarea densities. The exact information state is the full unnormalized belief. At zero evidence we use the real oriented blow-up of the positive cone and do not identify different limiting directions.

1.1. Positive observation disintegration. Let E be the compact weighted microstate/trajectory set selected by the B2 containment function, and let $\mu \in \mathcal{P}(E)$. A control u chooses a positive observation kernel $L_u(x, dy)$ on a Polish space Y . The joint prediction-observation measure is

$$J_{\mu,u}(dx, dy) = \mu(dx)L_u(x, dy).$$

Let $\nu_{\mu,u}$ be its Y -marginal and $\mu_{u,y}$ a Rokhlin conditional law:

$$J_{\mu,u}(dx, dy) = \mu_{u,y}(dx)\nu_{\mu,u}(dy).$$

Theorem 1.1 (Positive posterior kernel). *The map $(\mu, u) \mapsto J_{\mu,u}$ is positive and affine in μ . For every regular noiseless observation $O_u : E \rightarrow \mathbb{R}^d$ which is a submersion on a finite Whitney stratification, $\nu_{\mu,u}$ and $\mu_{u,y}$ have the coarea representation*

$$\nu_{\mu,u}(dy) = \sum_S \left[\int_{S \cap O_u^{-1}(y)} \frac{f_S(x)}{J_S O_u(x)} d\mathcal{H}^{\dim S - d}(x) \right] dy,$$

and $\mu_{u,y}$ is the normalized positive fiber measure. Oriented normal currents may encode the fiber geometry but are never paired with the scalar 1 to define evidence.

Proof. Existence of the disintegration is Rokhlin's theorem on standard Borel spaces. Positivity and affinity are immediate from the joint kernel. On a smooth stratum, apply the coarea formula to the positive density $f_S d\mathcal{H}^{\dim S}$. Lower-rank strata are included as separate positive components of the joint measure, with their own Hausdorff dimension. The evidence is the mass of these positive coefficient measures, not the mass or orientation of a current. $\square$

1.2. The blown-up zero-evidence cone. Let $\mathcal{K}_M \subset \mathcal{P}(E)$ be a compact moment sublevel. Define

$$\text{Bl}_0 \mathcal{M}_+(\mathcal{K}_M) = ((0, 1] \times \mathcal{K}_M) \cup (\{0\} \times \mathcal{K}_M).$$

The point (r, ν) represents the finite measure $r\nu$. At $r = 0$, different directions ν remain different. Give this space the product topology.

Lemma 1.2 (Compact positive evidence blow-up). *The blown-up cone is compact. The collapse map $(r, \nu) \mapsto r\nu$ is continuous but not injective at zero. A flat family with oscillating normalized direction has convergent*

subsequences in the blow-up precisely when its directions do; no fictitious canonical direction is assigned.

Proof. It is the compact product $[0, 1] \times \mathcal{K}_M$. The remaining statements follow directly from the definition. In particular the family $e^{-1/t^2} \nu_t$ is represented by $(e^{-1/t^2}, \nu_t)$, so its oscillation is visible rather than collapsed. $\square$

Prediction and observation are positively homogeneous on finite measures. For $\Lambda = r\nu$, let $P_u \Lambda$ be the predicted finite measure and let $O_{u,y} \Lambda$ be its positive fiber coefficient.

Theorem 1.3 (Feller belief transition). *Assume the prediction kernel is Wasserstein–Lipschitz and the regular observation densities satisfy, on $\mathcal{K}_M$,*

$$\|g_{\nu,u} - g_{\tilde{\nu},u}\|_{L^1(Y)} + \int g_{\nu,u}(y) W_1(\nu_{u,y}, \tilde{\nu}_{u,y}) dy \leq L W_1(\nu, \tilde{\nu}).$$

Then the prediction–observation transition extends continuously to the blown-up cone:

$$(r, \nu) \longmapsto \int \delta_{(rg_{\nu,u}(y), \nu_{u,y})} dy.$$

At zero evidence the direction evolves as an explicit state coordinate; it is not required to be determined by the zero measure alone.

Proof. Positive homogeneity gives the new mass $rg_{\nu,u}(y)$. The displayed estimate controls both evidence mass and normalized posterior in the product Wasserstein metric. Dominated convergence proves continuity for $r > 0$. If $r_n \rightarrow 0$ and $\nu_n \rightarrow \nu$, the mass coordinate tends to zero while the direction transition converges by the same estimate. Thus the extension is Feller on the product blow-up. $\square$

For the hard-sphere observation family, the estimate follows from the B1 inserted coefficient theorem below; for the Sinai family it follows from the A2 insertion LLT.

1.3. Exact full-belief dynamic programming. Let $b = (r, \nu)$ be the blown-up unnormalized belief. Prediction under a control and subsequent observation use the kernel of Theorem 1.3. Let running rewards be bounded and continuous.

Theorem 1.4 (Belief-state game). *For finite or continuous time, the exact partially observed risk-sensitive game is Markov on the complete state (physical state, b). Its Bellman operator is Feller, monotone, and cash additive. One-time microcanonical preparation, reward-only feedback, and adaptive canonical-law control remain three distinct games.*

Proof. Ionescu–Tulcea constructs the controlled joint law from the prediction and observation kernels. Conditional expectation of every future payoff depends on the observation history only through the current unnormalized posterior, including its zero-mass direction. Composition of the Feller kernels gives the Bellman recursion. Preparation changes the initial belief once;

a reward control changes only the running payoff; a canonical-law control changes the transition likelihood. Their strategy spaces and timing are therefore not interchangeable. $\square$

1.4. Inserted local coefficients. Let Y^ε be a finite vector of additive particle-path and actual-contact observations. Insert a bounded central belief test G in the B1 exact-number coefficient.

Theorem 1.5 (Uniform inserted coefficient). *On a compact regular source, control, and observation chart,*

$$\mathbb{E}[G; \sqrt{\mu_\varepsilon}(Y^\varepsilon - y) \in W_\varepsilon] = c_\varepsilon(y) \mathbb{E}_y^{\text{tilt}} G (1 + o(1)),$$

where $c_\varepsilon(y)$ is the B1 shell coefficient and the error is uniform for G in the unit ball of a fixed finite-history Lipschitz class. The denominator has the same coefficient, so the posterior ratio converges to the tilted conditional law.

Proof. Adjoin the observation source to the coefficient-level trajectory polymer gas of B1. Normal convergence holds uniformly over the compact control chart. The central spectral coefficient with insertion is obtained by differentiating the same exact- N Cauchy integral; the left and right coefficient eigenvectors pair with G . The compact annulus and speed-dependent high-frequency estimates are unchanged because G affects only finitely many history coordinates. Fourier inversion gives the same Gaussian coefficient times the tilted expectation of G . $\square$

This theorem proves the quantitative observation estimate assumed in Theorem 1.3 by duality over bounded Lipschitz tests.

1.5. Growing finite belief coordinates. A fixed finite list of moments does not determine a probability law. Let $\mathcal{K}_M$ be compact in W_1 . For $\epsilon > 0$, choose a finite ϵ -net $\{\phi_1, \dots, \phi_{K(\epsilon)}\}$ in the unit ball of bounded Lipschitz functions restricted to $\mathcal{K}_M$, and define

$$C_\epsilon(\nu) = (\nu(\phi_1), \dots, \nu(\phi_{K(\epsilon)})).$$

Choose a measurable nearest-point reconstruction $R_\epsilon(C_\epsilon \nu) \in \mathcal{K}_M$.

Lemma 1.6 (Quantitative growing-coordinate reconstruction). *Uniformly on $\mathcal{K}_M$,*

$$W_1(\nu, R_\epsilon C_\epsilon \nu) \leq 3\epsilon.$$

If the belief kernel has Lipschitz constant L , the N -stage value computed with reconstructed beliefs differs from the full-belief value by at most

$$3\epsilon L_V \sum_{j=0}^{N-1} L^j.$$

Thus choosing $\epsilon = \epsilon_\varepsilon \downarrow 0$, equivalently $K_\varepsilon \rightarrow \infty$, gives a uniform $o(1)$ reduction on every fixed horizon.

Proof. By Kantorovich duality, W_1 is the supremum over the bounded-Lipschitz unit ball. Approximate the maximizing test by one of the finite net functions and use the nearest-coordinate reconstruction; the three approximation errors give 3ϵ . Couple the exact and reconstructed belief chains at each step. The kernel Lipschitz estimate propagates the current error by L ; the value Lipschitz constant L_V and telescoping give the displayed sum. This constructs a reduced Markov kernel by pushforward through C_ϵ and R_ϵ , rather than treating moment proximity as an exact state. $\square$

1.6. A fully specified local statistical limit. Let $\theta \in \Theta \subset \mathbb{R}^d$ parametrize a regular prepared source. Assume a prior density positive and C^1 near the true θ_0 , uniform identifiability away from θ_0 , and the exact finite-centred LAN expansion

$$\log \frac{dP_{\theta_0+h/\sqrt{\mu}}^\epsilon}{dP_{\theta_0}^\epsilon} = h \cdot Z_\epsilon - \frac{1}{2} h^T I_{\theta_0} h + o_{P_{\theta_0}}(1),$$

uniformly for h in compact sets. Here Z_ϵ is the B3 joint density/contact score and $I_{\theta_0} > 0$.

Theorem 1.7 (Controlled Bernstein–von Mises theorem). *On every compact informative strategy class for which the inserted coefficient and LAN expansion are uniform, the posterior law of $\sqrt{\mu_\epsilon}(\theta - \theta_0)$ converges in total variation to*

$$N(I_{\theta_0}^{-1}Z, I_{\theta_0}^{-1}).$$

The convergence is uniform over the strategy class.

Proof. The inserted coefficient gives a common local likelihood normalization and the B3 process theorem gives $Z_\epsilon \Rightarrow N(0, I_{\theta_0})$. Uniform identifiability makes the posterior outside a fixed local neighborhood exponentially small. On compact h -sets, uniform LAN and the C^1 prior reduce numerator and denominator to the Gaussian integral. Tail tightness follows from the positive information matrix. Scheffé’s lemma gives total variation convergence uniformly in the strategy. $\square$

Theorem 1.8 (Round-fourteen C1 closure). *The exact information state is a positive disintegrated full belief on a compact blown-up evidence cone; the belief game is Feller; local observations have an inserted coefficient theorem; growing finite coordinates approximate the full game quantitatively; and the regular controlled statistical model satisfies the stated Bernstein–von Mises limit.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, and 1.7, with Lemma 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and

build gates are internal reproducibility checks and do not replace independent external mathematical review.

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